NAICHENG JIANG

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RESEARCH INTERESTS

3D generative modelling and spatial representations; vision-language understanding; reliable multimodal AI, including applications in health.

EDUCATION

3+2 bachelor's–master's study pathway.

KTH Royal Institute of Technology, Stockholm, Sweden

MSc in Information and Network Engineering | Aug 2025 – Jun 2027 (expected)

Weighted grade average: **4.72/5.00** (self-calculated from completed graded courses).

Shanghai Jiao Tong University, Shanghai, China

BSc in Electrical and Computer Engineering | Aug 2022 – Aug 2026

UM-SJTU Joint Institute (Global College) | GPA: 3.57/4.00

HONORS AND AWARDS

Outstanding Graduate Award, Undergraduate Class of 2026 — Shanghai Jiao Tong University

Undergraduate Academic Scholarship, Class C, Shanghai Jiao Tong University (2023 and 2024)

Yu Liming Scholarship, UM-SJTU Joint Institute (2024)

RESEARCH EXPERIENCE

VecSetX — Compositional 3D Generation from Incomplete Supervision

Co-first author; manuscript in preparation for CVPR 2027 | Feb 2026 – Present

Conditional completion and unconditional generation of 3D anatomy from four partially annotated datasets.

- Extended and fine-tuned a **multi-source 3D tokenizer** for latent generative modelling, achieving 0.923 mean reconstruction Dice across 204 validation cases.
- Prototyped a **conditional diffusion transformer** with cross-attention; designed context-shuffling controls to test evidence use and diagnose conditioning failures.
- Evaluated **cross-source generalization** and geometric consistency through baseline comparisons, distinguishing structural plausibility from patient-specific accuracy.

CURRENT AND PLANNED PROJECTS

Flow Matching for Image Reconstruction and Translation | KTH EQ2443

Ongoing course project | Oct 2026 – Present

- Designing a U-Net velocity-field baseline for flow-matching reconstruction from perturbed images, with image-to-image translation as a further research objective.
- Planning controlled evaluations of reconstruction fidelity, flow-estimation accuracy and sampling cost, using image-quality metrics and runtime to study quality–compute trade-offs.

3D Vision–Language Understanding | Proposed research

Proposed research on grounding language-based reasoning in volumetric visual evidence, with emphasis on spatial relationships and consistency between generated responses and observed anatomy.

TECHNICAL SKILLS

Python, MATLAB, C++; PyTorch, OpenCV, NumPy/SciPy; Linux, Git, LaTeX, Slurm, LUMI/HPC, CUDA.

LANGUAGES

Chinese (native); English (professional working proficiency).